



VL-FPGA-B

BASIC FPGATRainer

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



VLSI Training System is the first of its kind using actual techniques of VLSI. It is our vision to furnish products, which are reliable and convenient to use. AKADEMIKA offers a high degree of expertise in developing efficient products.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

•Design• Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

Important Information

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/ underperforms if it is not operated under following conditions:
 - Ambient temperature: Between 0 to 45 ° C
 - Relative humidity: Between 20 to 80 %
- Avoid any immediate/significant change of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service centre or the sales outlet.

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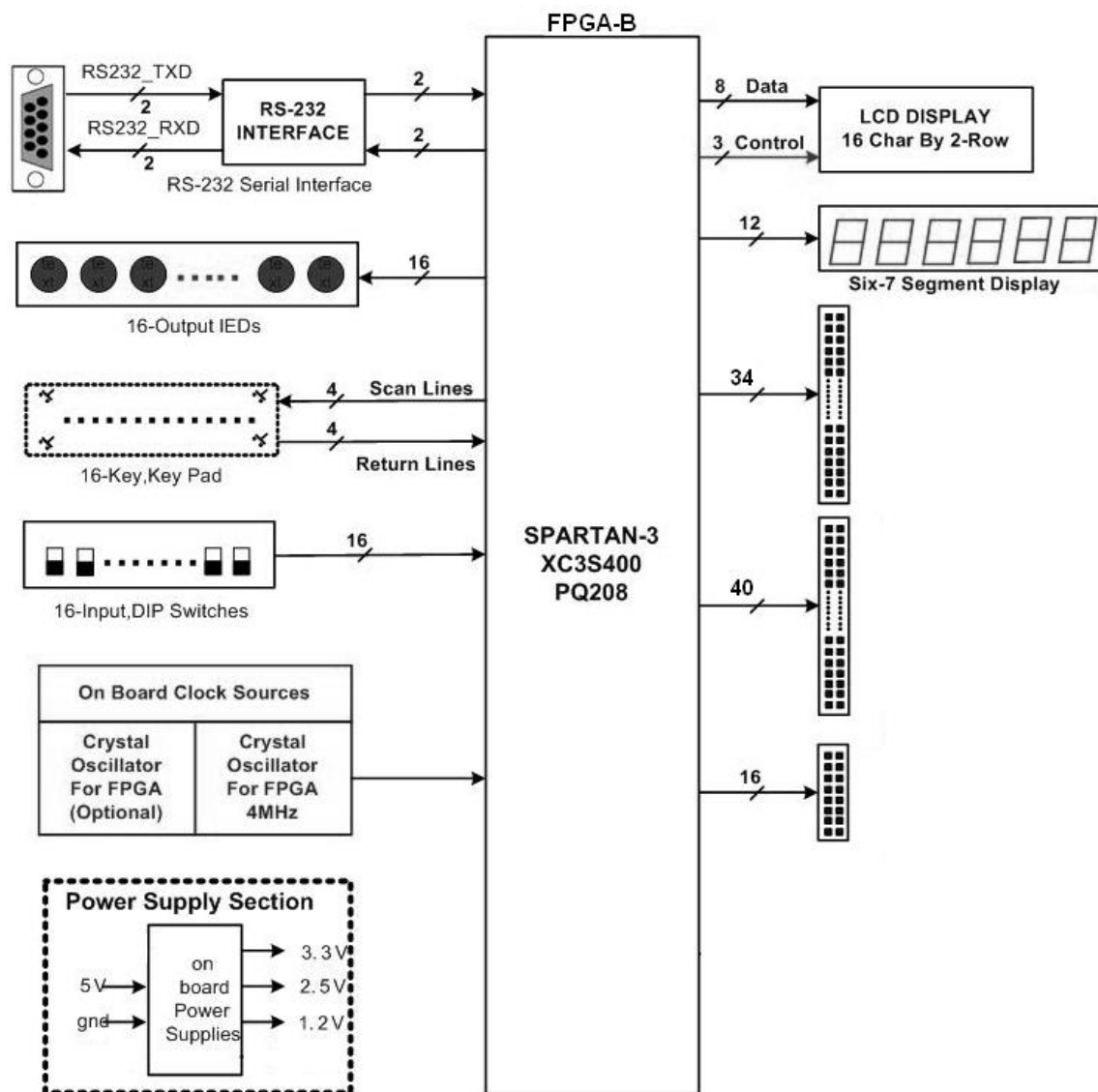
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TECHNICAL SPECIFICATIONS

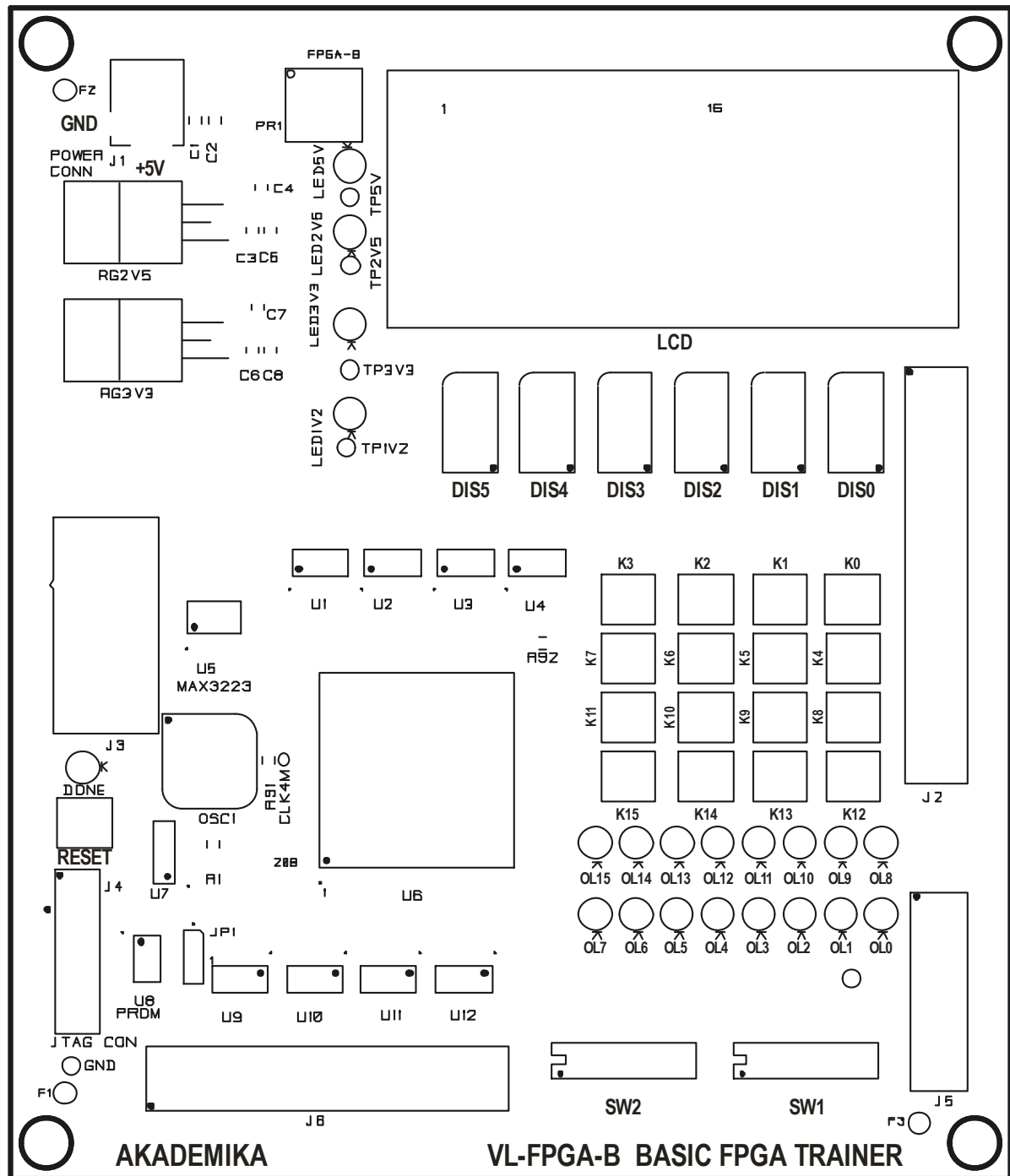
- **SPARTAN -3 FPGA :** 400 K logic cell SPARTAN -3 FPGA in PQ208 Plastic Quad Flat Package
- **DIP Switches:** 16 DIP switches.
- **LEDs:** 21 onboard LEDs
 - 16 output LEDs (OL0 – OL15).
 - Done LED.(DONE)
 - 4 Power ON LEDs (LED5V, LED3V3, LED2V5, LED1V2).
- IOs are available on connector.
- **Push Button Switches:** 16 momentary-contact push button switches in 4x4 matrix.
- **Display:**
 - i) One 16X4 LCD display.
 - ii) Six multiplexed common anode seven segment displays
- On board 4 MHz clock and Power On reset circuit.
- **POWER SUPPLY.**
Adapter is supplied with VL-FPGA-B Trainer. Required supply voltage for FPGA is generated on board.
- **TEST POINTS [TPS]**
User can use these points to verify 5V, Ground and Clock
- **PROGRAMMING:**
Programming the device through USB JTAG cable.
- **Deliverables :**

JTAG Cable	: 01
Manual	: 01
5v DC Adapter_1.5 A	: 01
PenDrive	: 01

Functional Block Diagram



VL-FPGA-B COMPONENT SIDE



INTRODUCTION

BASICS ABOUT CPLD TRAINER

The FPGA Trainer **VL-FPGA-B** is useful to realize and verify various digital designs. User can construct VHDL/Verilog code and verify the results by implementing physically into the target device FPGA. With the help of this trainer user can simulate/observe various input and output conditions to verify the implemented design. Also user can select various I/O standard interfaces to the device.

PROGRAMMABLE LOGIC DEVICES [PLDs]

A Programmable Logic Device is a device whose logic characteristics can be changed and manipulated or stored through programming.

DIFFERENT TYPES OF PLD'S.

PROGRAMMABLE ARRAY LOGIC [PALs]

The most common and simple device that falls in this category is the PAL, which simply consists of an array of AND gates and an array of OR gates. The AND array is programmable while the OR array is relatively fixed.

FIELD PROGRAMMABLE GATE ARRAYS [FPGAs]

FPGA's are arrays of logic blocks, which can be linked together to form complex logic implementations. They are separated into two categories Fine Grained and Coarse Grained.

Fine Grained being made up of sea of gates or transistors or small macro cells, while Coarse Grained being made up of bigger macro cells which are often made up of flip-flops and Look up Tables which make up the Combinational logic functions. These are **RAM based devices** i.e. these devices lose their configuration when power is switched off. Hence they have to be configured every time when power is applied.

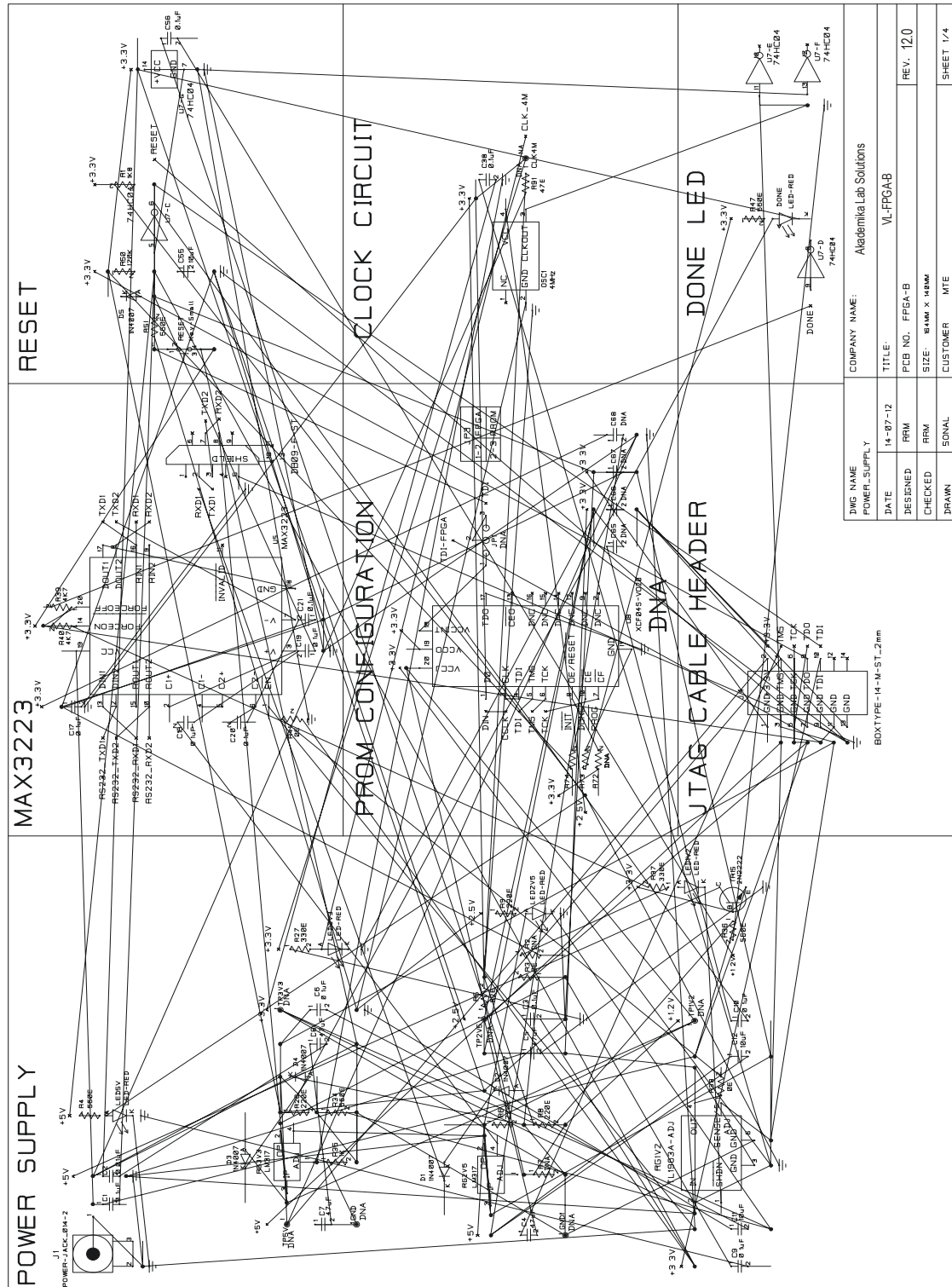
COMPLEX PROGRAMMABLE LOGIC DEVICES [CPLDs]

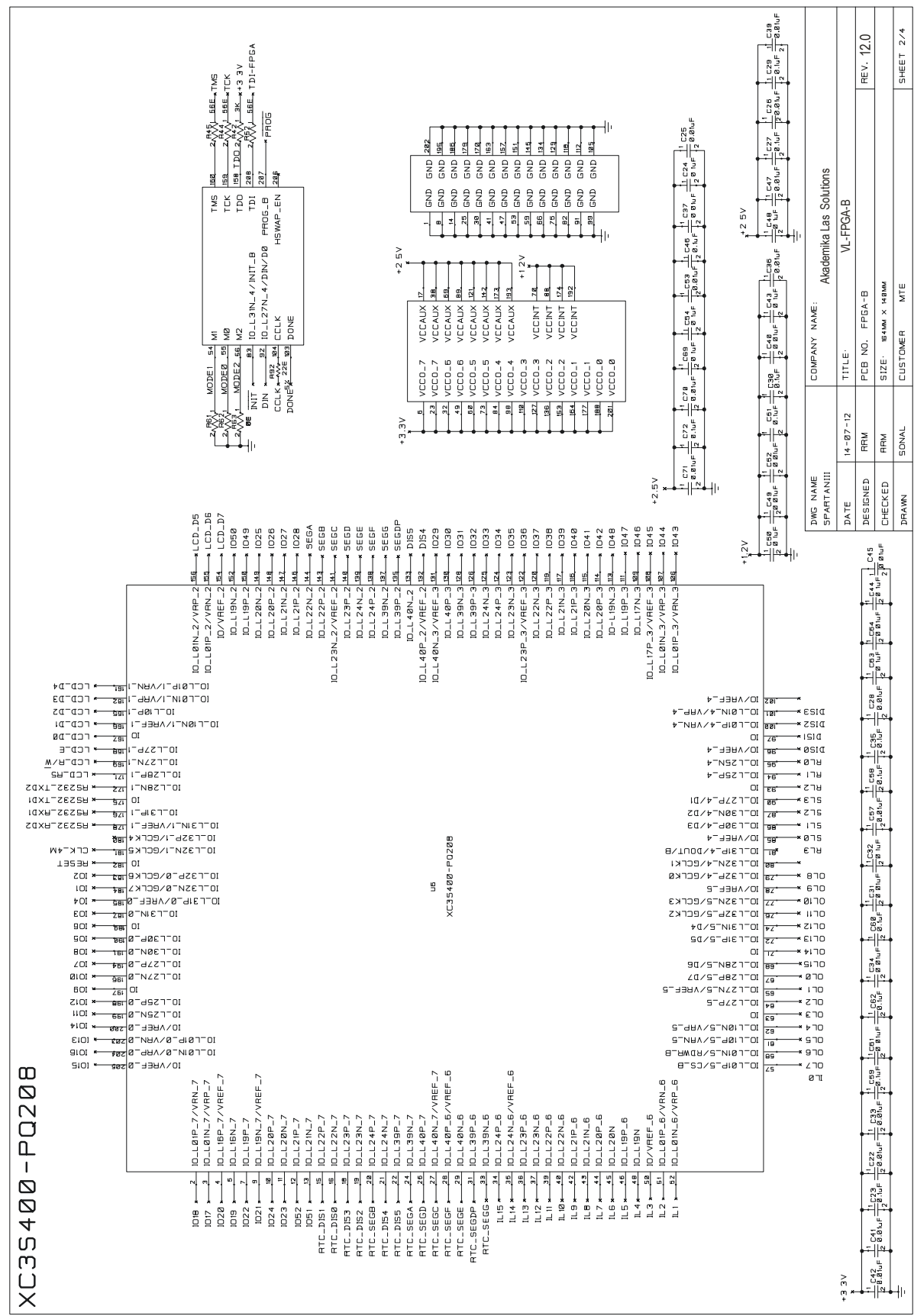
CPLD's are made up of smaller common Macro cells, which are programmable. CPLD's consists of multiple PAL like function block that can be interconnected through a switch matrix. These are **[Flash] EPROM based devices** i.e. these devices store their configuration even when power is switched off. Hence they need not to be configured every time when power is applied.

APPLICATION SPECIFIC INTEGRATED CIRCUITS [ASICs]

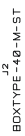
ASIC's are nothing but prefabricated pre-doped silicon chips. These are application specific designs. They cannot be reconfigured once manufactured. Once the design is completely finalized, it can be made as ASIC. Design changes are not possible but the size and speed is more.

VL-FPGA-B Circuit Diagram

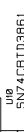
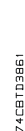
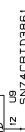




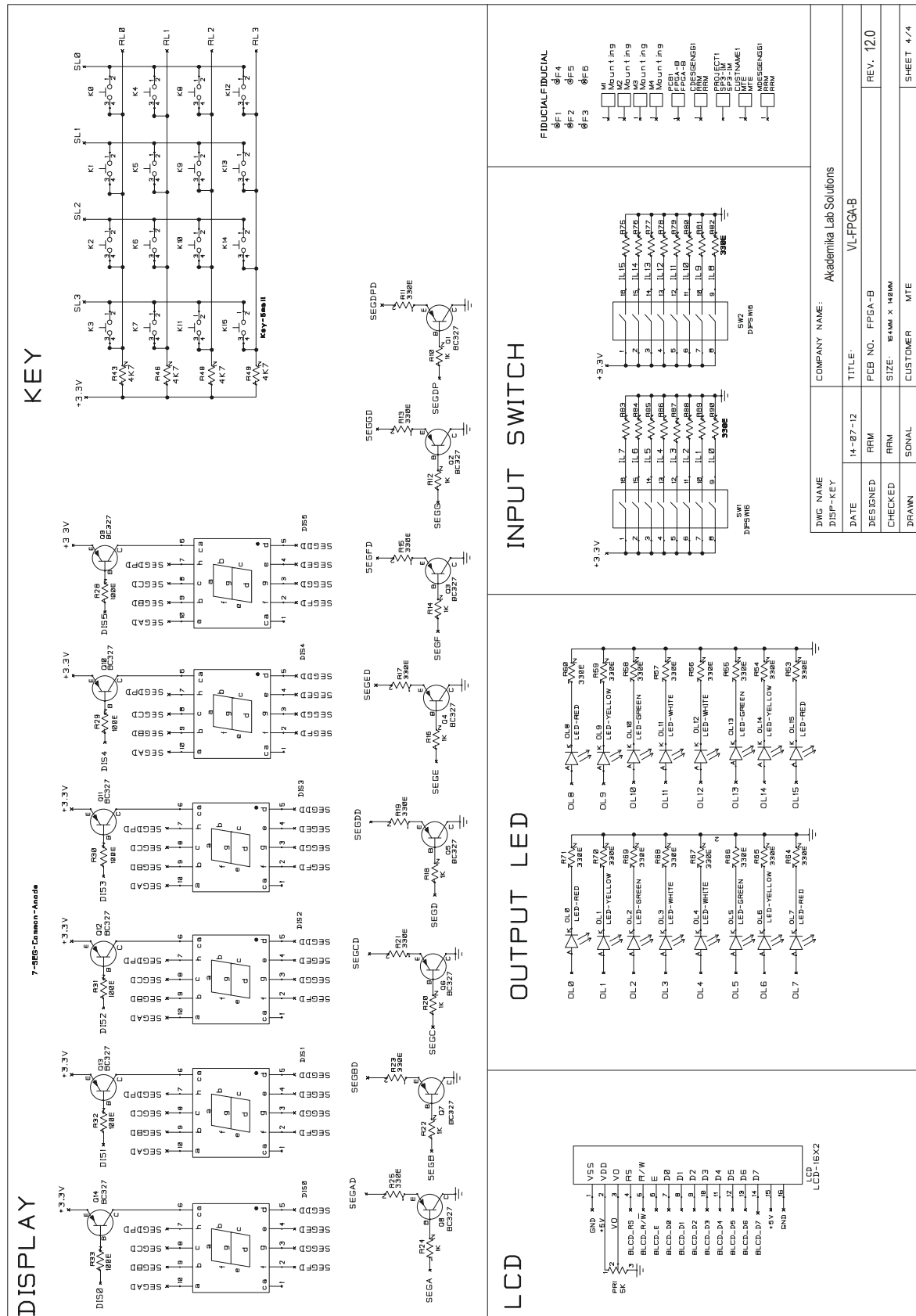
AKADEMIKA



QUICK SWITCHS



DWG NAME	COMPANY NAME: Akademika Lab Solutions		
JO CONNECTOR			
DATE	14-07-12	TITLE:	VL-FPGA-B
DESIGNED	ARM	PCB NO.	FPGA-B
CHECKED	ARM	SIZE:	154MM x 148MM
DRAWN	SIGNAL	CUSTOMER	MTE
			SHEET 3/4



EXPERIMENT NO.1

EXPERIMENT NO. 1

NAME

Implementation and verification of 8:1 Multiplexer in Xilinx Spartan-3 FPGA using Xilinx ISE 13.4 tool.

OBJECTIVE

To write the VHDL code for 8:1 MUX and verify the result on output led and 7-segment display.

EQUIPMENTS

- VL-FPGA-B board.
- Power supply.
- J-Tag Cable.

THEORY

The Multiplexer is a combinational circuit that is the most widely used circuit in Digital Design. Multiplexing means transmitting a large number of information units over a smaller number of channels or lines.

A Digital Multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. The selection of a particular input line is controlled by a set of selection lines.

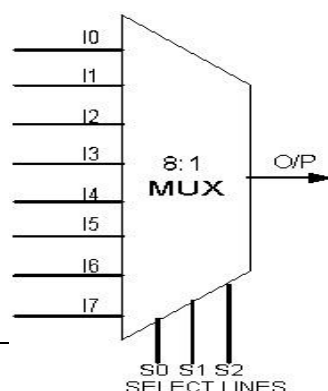
DESIGN DESCRIPTION

For selecting one out of 'n' inputs for connection to the output, a set of m select inputs are required.

$$2^m = n$$

(n= Number of inputs, m = Number of select lines)

Depending upon the digital code applied at the select inputs one out of 8 data sources is selected & transmitted to a single output Channel.



Select Inputs			Output
S0	S1	S2	Y
0	0	0	Sw11
0	0	1	Sw12
0	1	0	Sw13
0	1	1	Sw14
1	0	0	Sw15
1	0	1	Sw16
1	1	0	Sw17
1	1	1	Sw18

Truth Table of 8:1 MUX

VHDL CODE FOR 8:1 MUX:

```

LIBRARY IEEE;
USE IEEE.STD_LOGIC_1164.ALL;
USE IEEE.STD_LOGIC_ARITH.ALL;
USE IEEE.STD_LOGIC_UNSIGNED.ALL;

```

```

ENTITY MUX_TEST IS
    PORT ( SW110 : IN  STD_LOGIC;
          SW12  : IN  STD_LOGIC;
          SW13  : IN  STD_LOGIC;
          SW14  : IN  STD_LOGIC;
          SW15  : IN  STD_LOGIC;
          SW16  : IN  STD_LOGIC;
          S17   : IN  STD_LOGIC;
          S0    : IN  STD_LOGIC;
          S1    : IN  STD_LOGIC;
          S2    : IN  STD_LOGIC;
          LED   : OUT STD_LOGIC);

```

```

END MUX_TEST;

```

```

ARCHITECTURE BEHAVIORAL OF MUX_TEST IS

```

```

    SIGNAL SEL : STD_LOGIC_VECTOR ( 2 DOWNT0 0);

```

```

BEGIN

```

```

    SEL <= S0 & S1 & S2; --- S0= MSB

```

```
LED <= SW11 WHEN SEL = "000" ELSE
    SW12 WHEN SEL = "001" ELSE
        SW13 WHEN SEL = "010" ELSE
            SW14 WHEN SEL = "011" ELSE
                SW15 WHEN SEL = "100" ELSE
                    SW16 WHEN SEL = "101" ELSE
                        SW17 WHEN SEL = "110" ELSE
                            SW18 ;
```

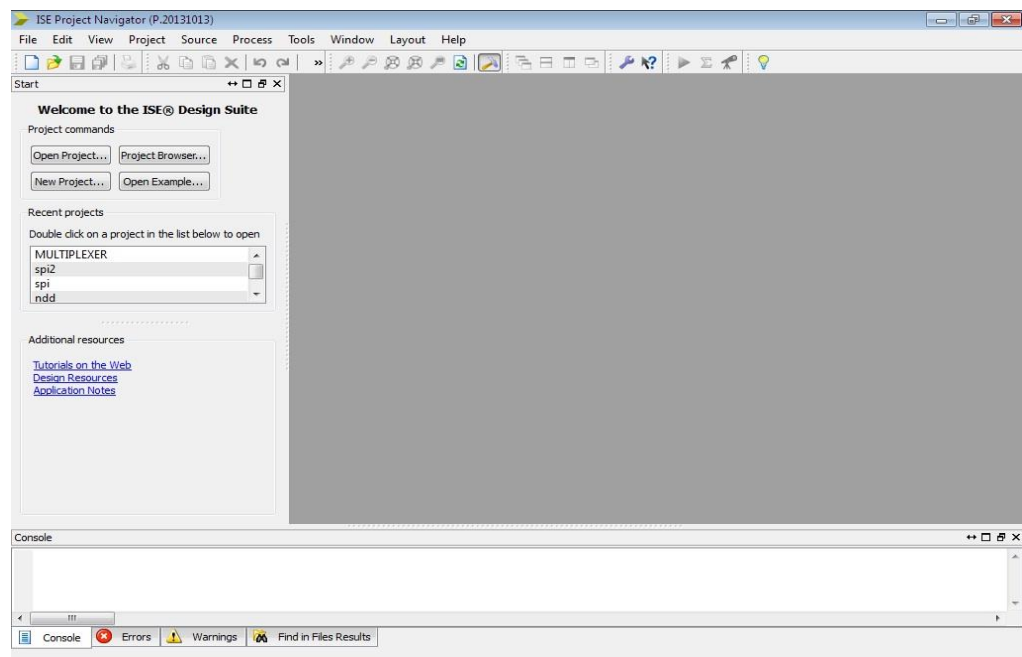
```
END BEHAVIORAL;
```

USER CONSTRAINT:

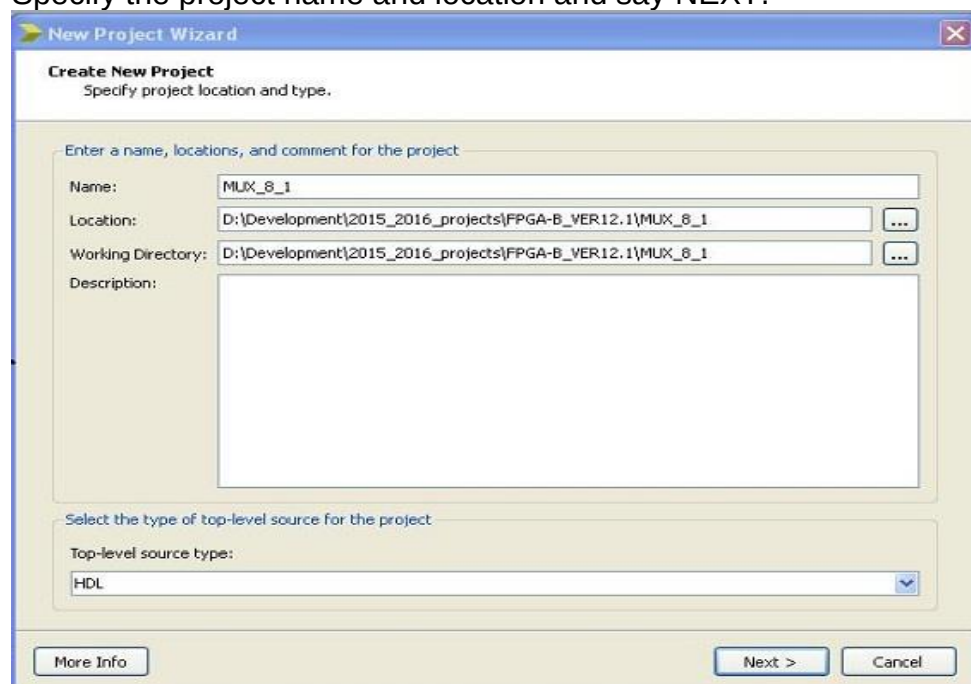
```
NET "SW11" LOC = "P44";
NET "SW12" LOC = "P45";
NET "SW13" LOC = "P46";
NET "SW14" LOC = "P48";
NET "SW15" LOC = "P50";
NET "SW16" LOC = "P51";
NET "SW17" LOC = "P52";
NET "SW18" LOC = "P57";
NET "S0" LOC = "P34";
NET "S1" LOC = "P35";
NET "S2" LOC = "P36";
NET "LED" LOC = "P68";
```

PROCEDURE:**STEPS TO IMPLEMENT THE 8:1 MUX IN FPGA USING XILINX ISE 13.4**

Step 1 : Start the Xilinx Project Navigator by using the desktop shortcut or by using the *Start → Programs → Xilinx ISE design suite(13.4)*



Create a new project. In the window go to FILE → New project.
Step 2 : Specify the project name and location and say NEXT.



Step 3: Select Device. Use the pull-down arrow to select the Value for each Property Name. Click in the field to access the pull-down list.

New Project Wizard

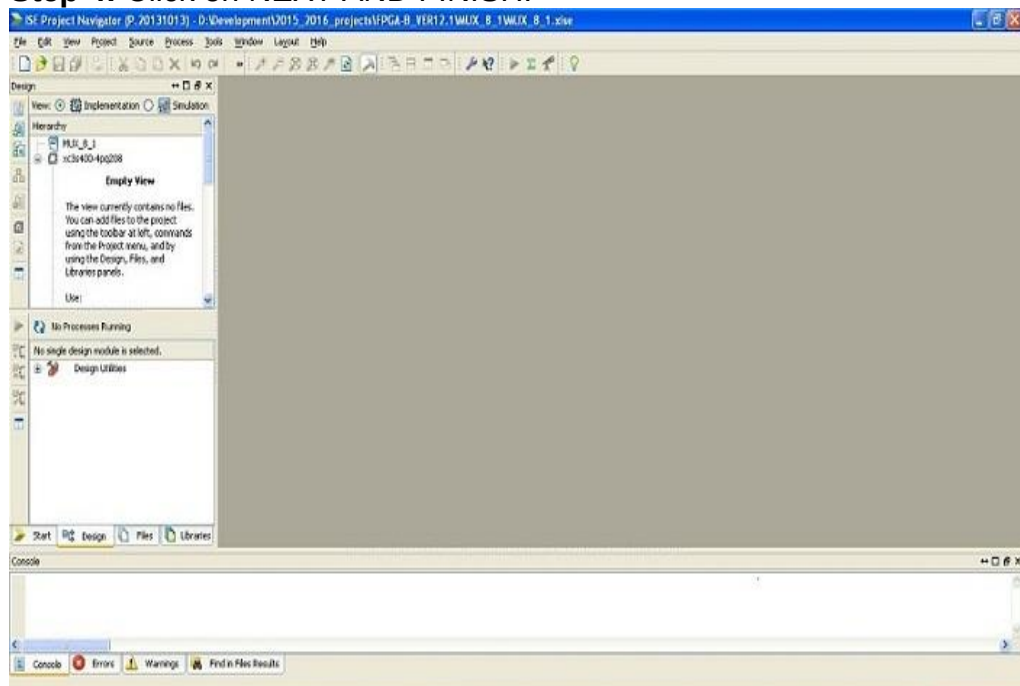
Project Settings
Specify device and project properties.

Select the device and design flow for the project

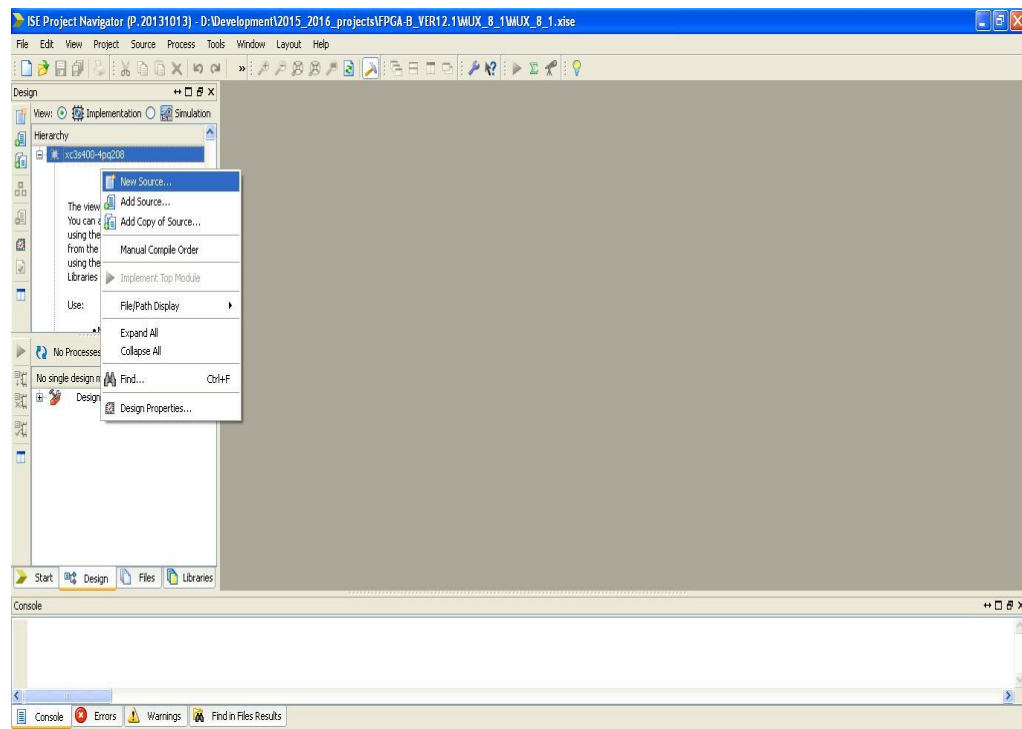
Property Name	Value
Evaluation Development Board	None Specified
Product Category	All
Family	Spartan3
Device	XC3S400
Package	PQ208
Speed	-4
Top-Level Source Type	HDL
Synthesis Tool	XST (VHDL/Verilog)
Simulator	ISim (VHDL/Verilog)
Preferred Language	VHDL
Property Specification in Project File	Store all values
Manual Compile Order	☐
VHDL Source Analysis Standard	VHDL-93
Enable Message Filtering	☐

More Info < Back Next > Cancel

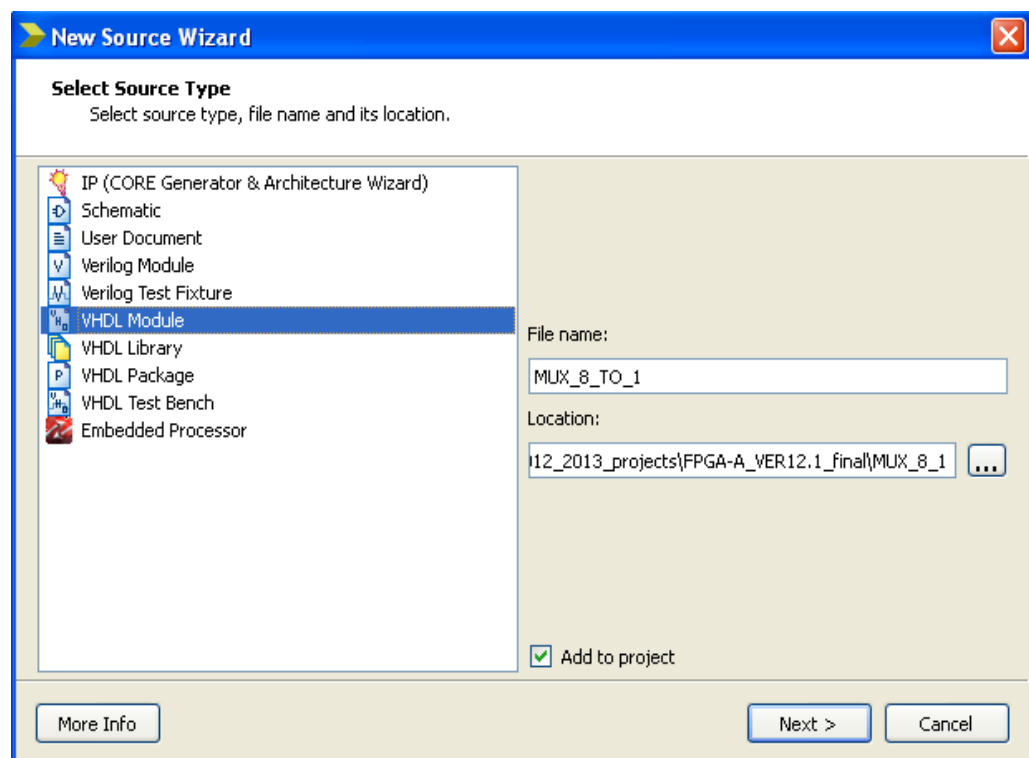
Step 4: Click on NEXT AND FINISH.



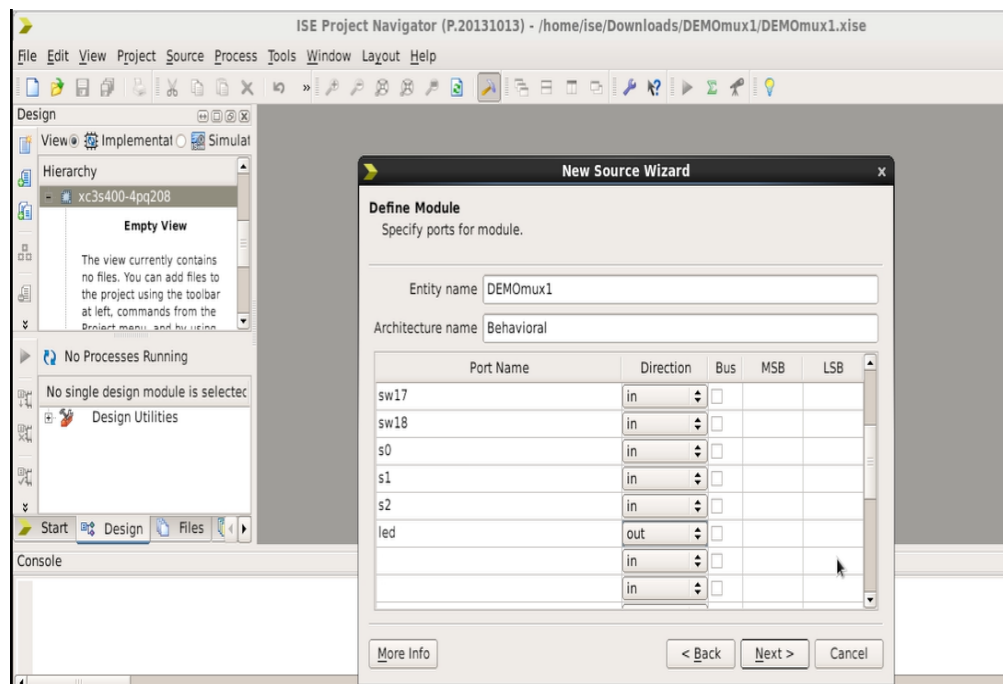
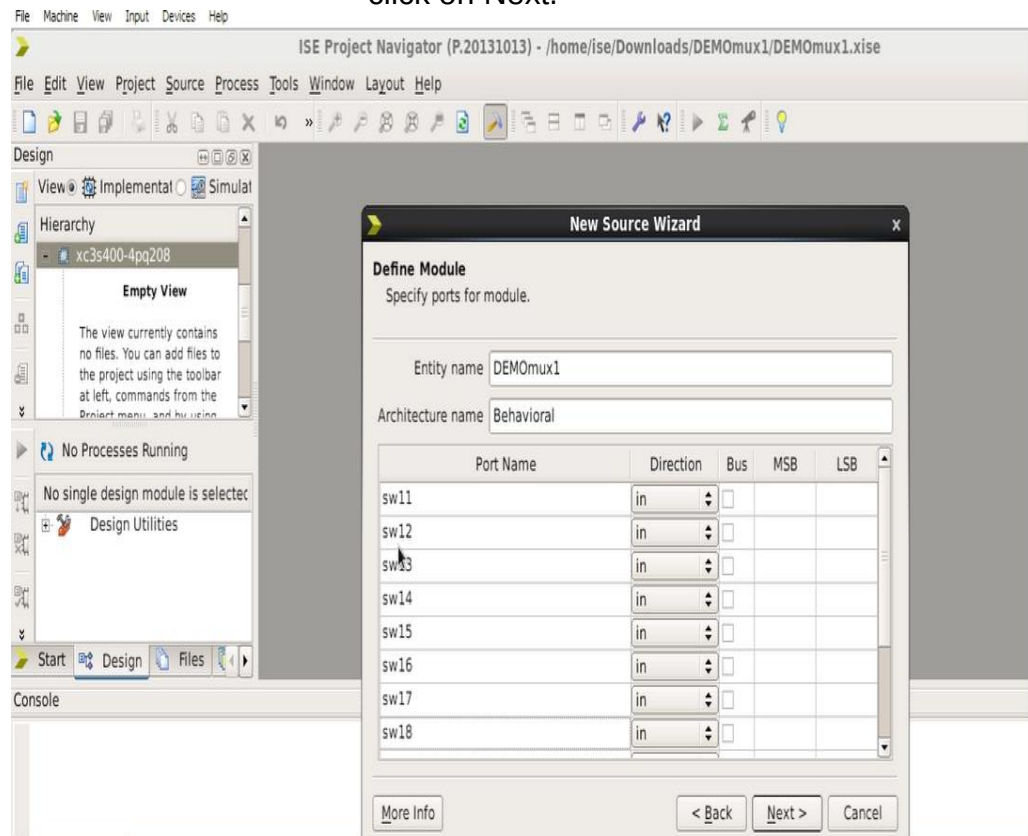
Step 5: Right click the mouse button on device and select New source.

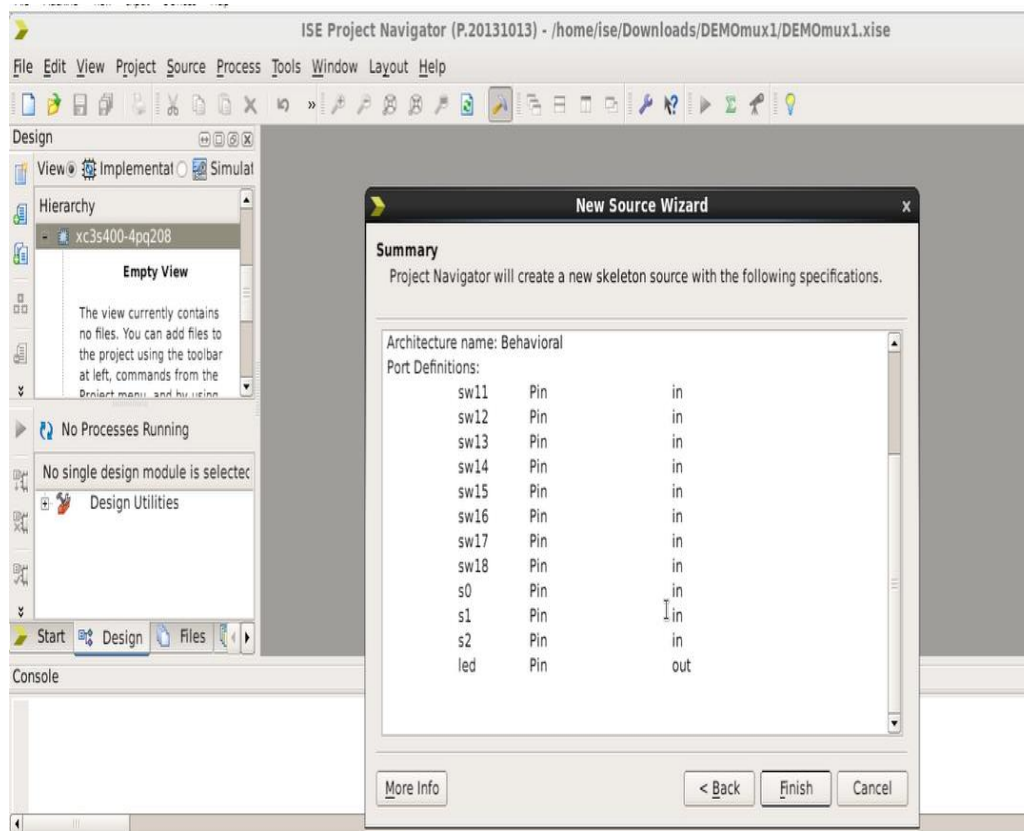


Step 6: Select VHDL module, give file name and click on Next.

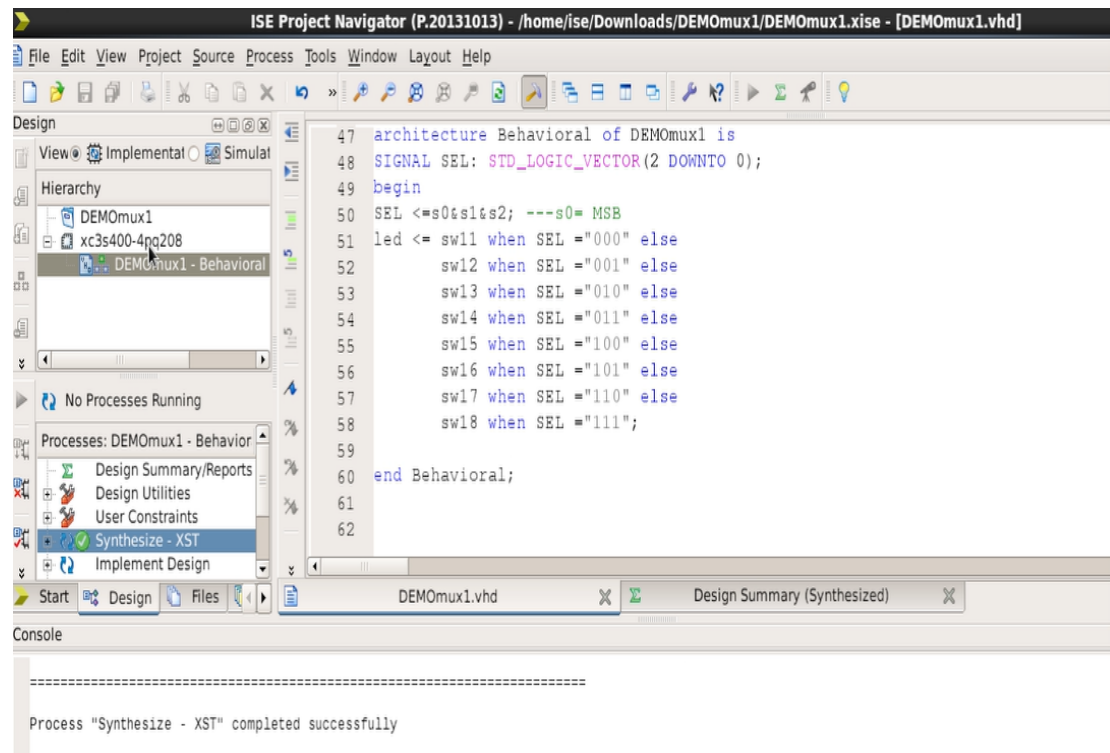


Step 7: Enter IO port name and their directions in respective fields and click on Next.



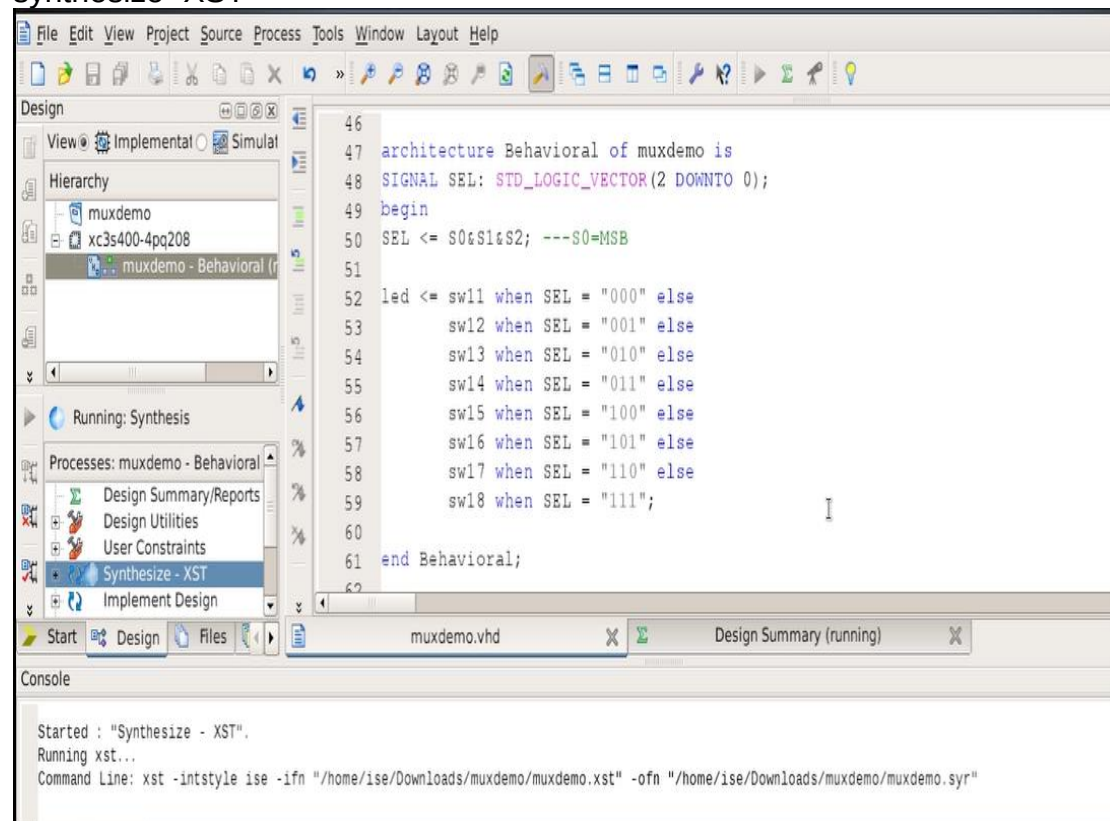


Step 8: Verify the design summary and click on finish. The empty workspace with library and entity will open.

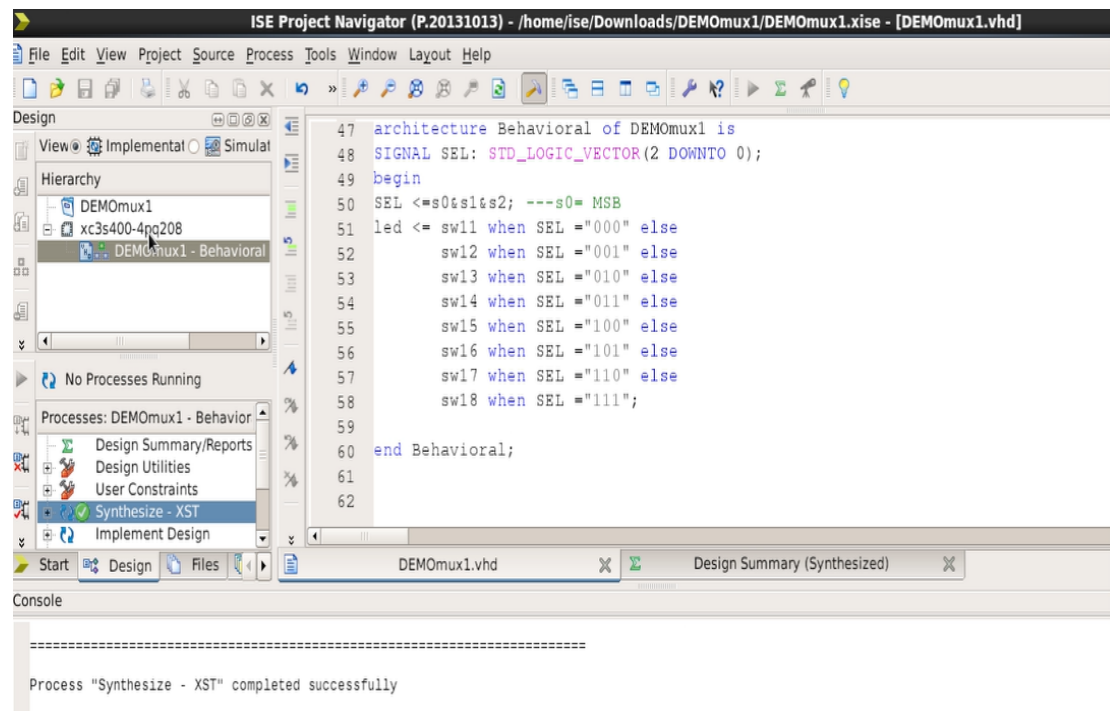


Step 9: Write your program after begin keyword and click on save.

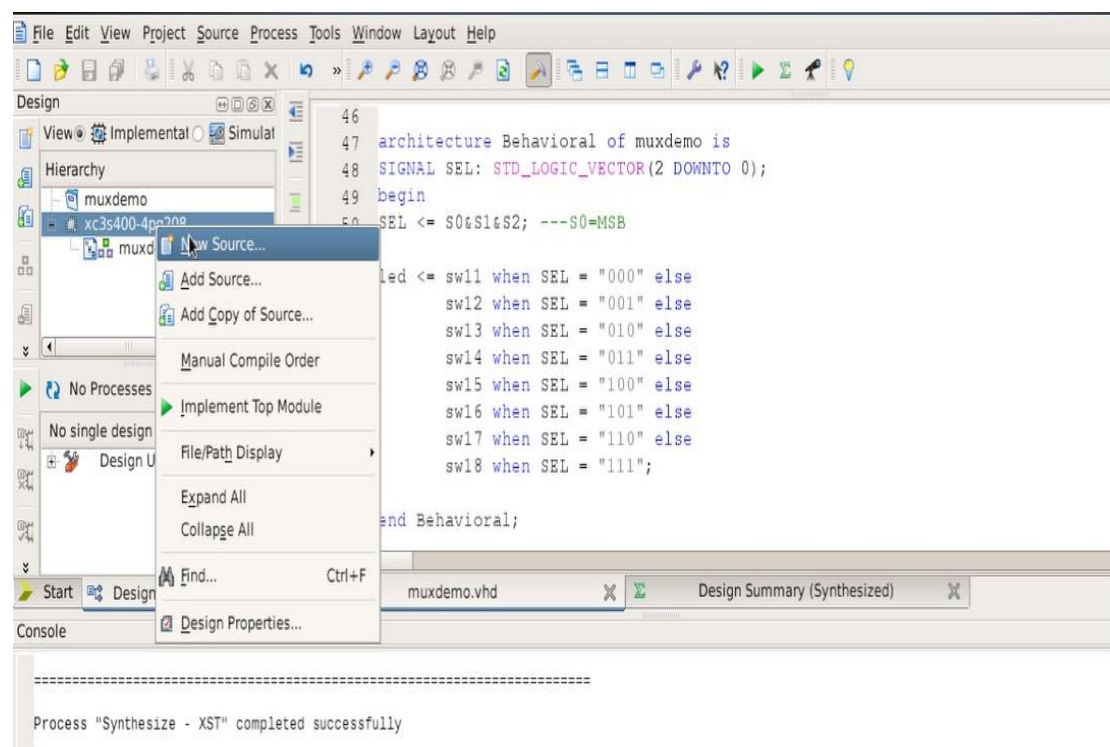
Step 10: To synthesize go to implement design and double click on synthesize- XST



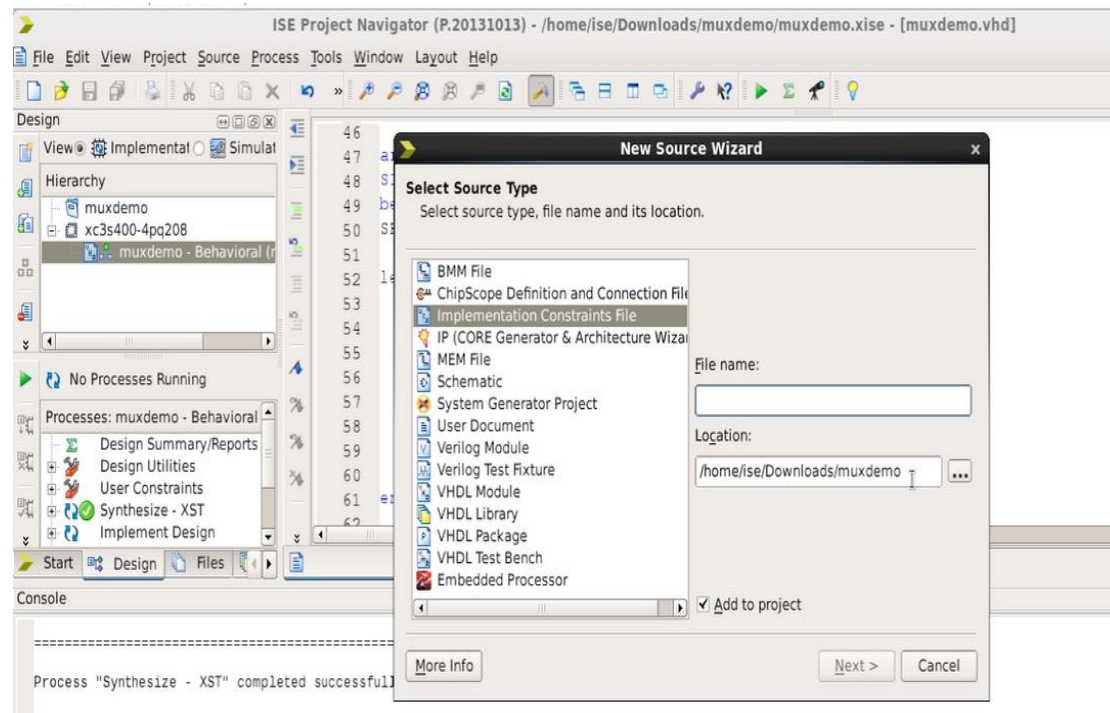
Step 11: If no error present in program console window gives message that Process "Synthesize - XST" completed successfully.



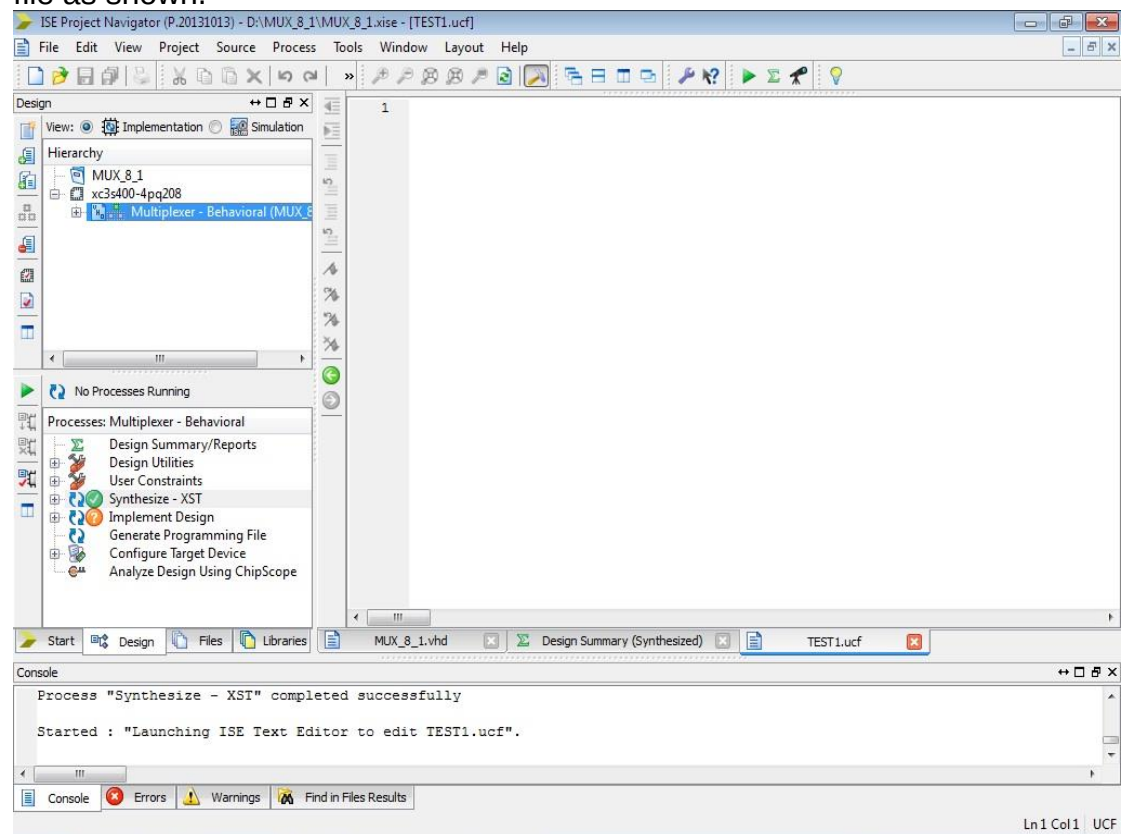
Step 12: Right click on device select New Source.



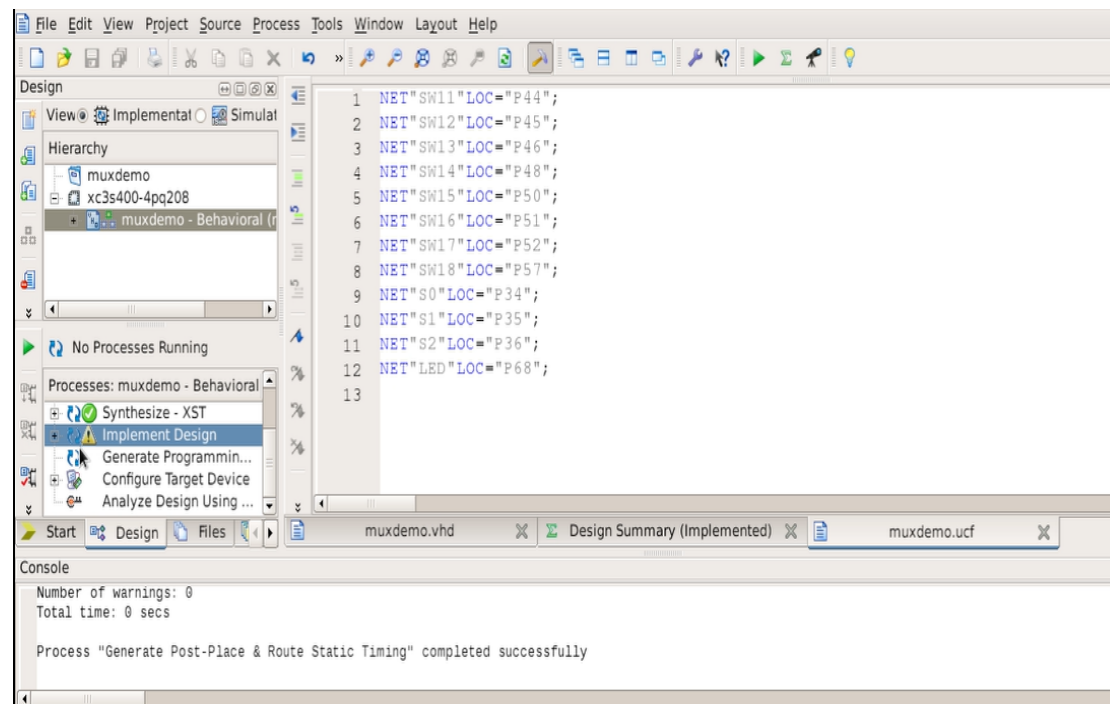
Step 13: Select Implementation Constraint file, enter file name and click next and finish.



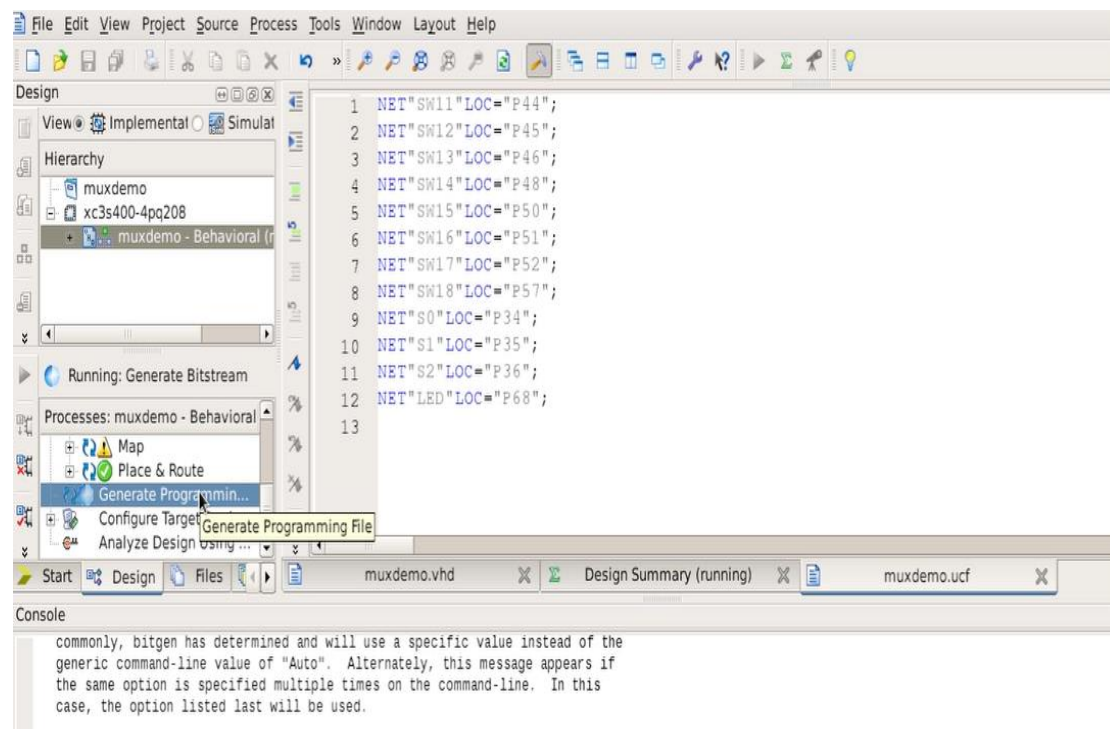
Step 14: An empty constraint file will be created and saves under main .VHD file as shown.



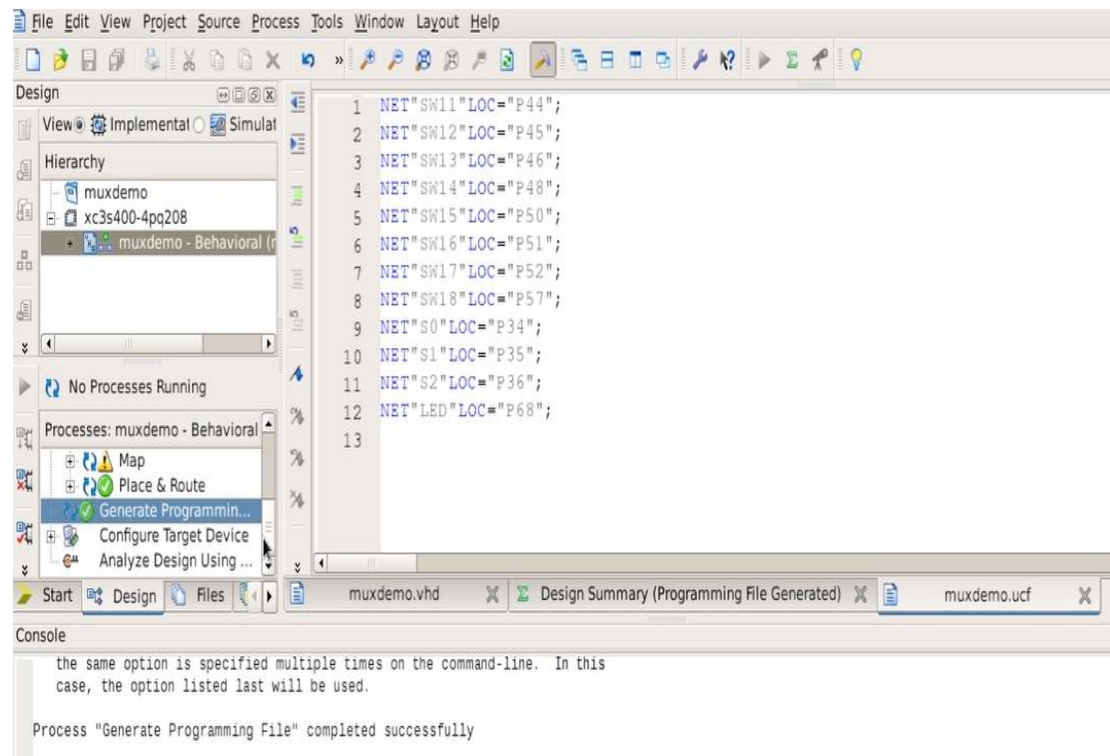
Step 15: Apply the input and output constraint by referring the Pin Assignment chapter. Click on save.



Step 16: To generate the programming file double click on generate programming file.



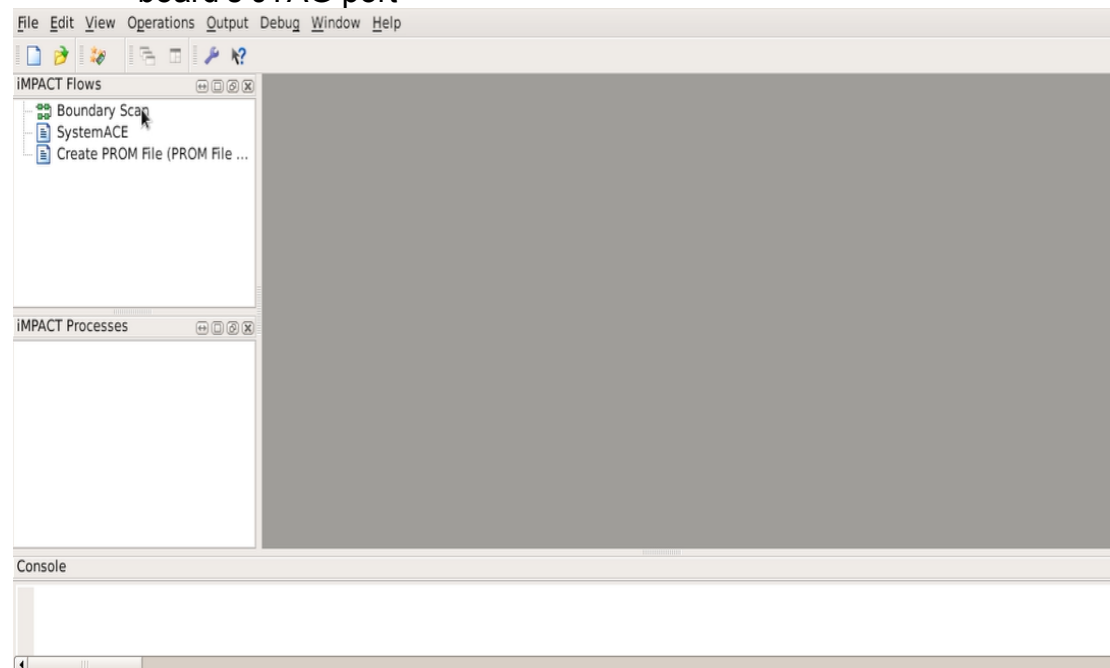
Step 17: Once all the processes are completed successfully the console window gives message as Process "Generate Programming File" completed successfully.



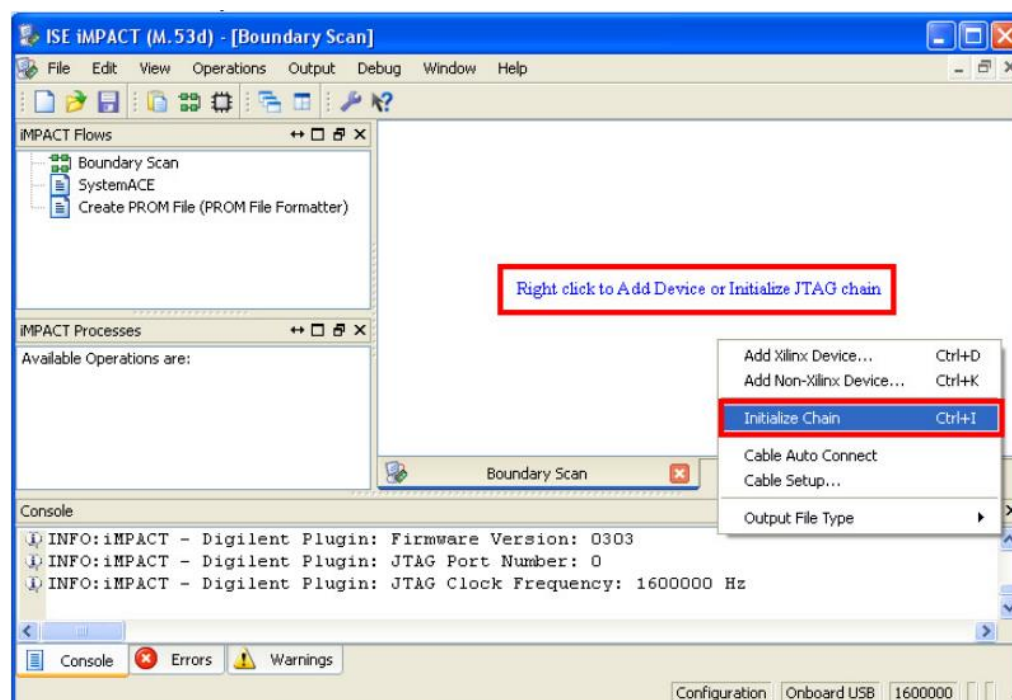
Step 18: Click on Configure Target device, It will Open Impact Automatically.

Step19: Program downloading procedure to target board.

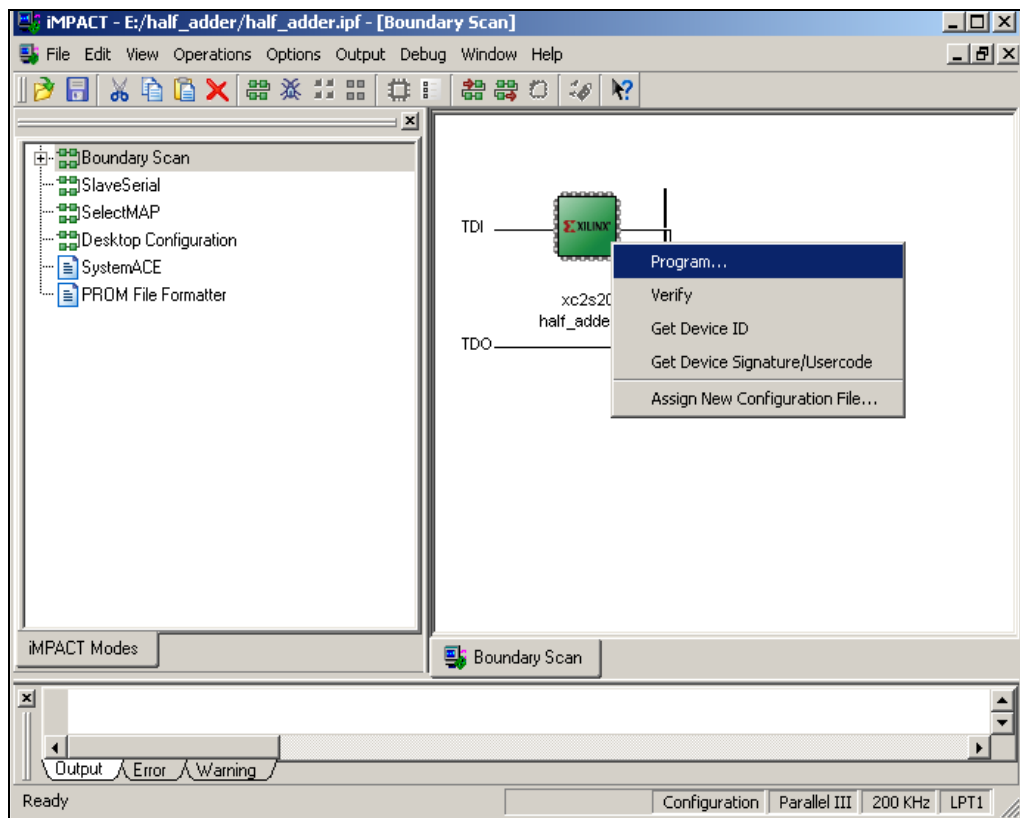
Connect USB JTAG cable provided to computer and other end to board's JTAG port



Right Click in the “Boundary Scan” window to “Initialize Chain”



The device is detected, give path for programming file. Right click on the device and click on *Program*.



If the device is programmed properly, it says Programming Succeeded or else. Programming Failed.

PROCEDURE:

1. The following files are generated which is useful for the design and implementation of 8:1 MUX.

mux_test.bit	Bit file
Test1.ucf	UCF file
Mux_test.vhd	VHDL file

2. Download multiplexer.bit. file into FPGA, when the device is successfully configured
3. The output is mapped on the LED's (OL0).
4. Each switch from sw1_1 to sw1_8 represents one input.
5. Sw2_1,sw2_2,sw3_3 represents the 3-bit select line.

EXPERIMENT NO.2

EXPERIMENT NO.2

NAME

To check the IO lines of FPGA

OBJECTIVE

To write the program to generate pulse train on FPGA IOs

EQUIPMENTS

- VL-FPGA-B board,
- Power Supply,
- DSO.

DESCRIPTION:

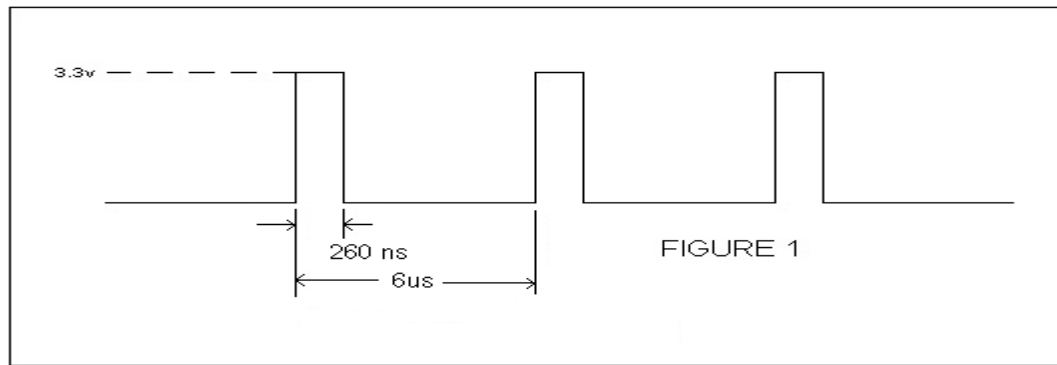
In this test/demo code we design a Shift register of 24 **Bit** which is shifting by one bit and its output is mapped on the free IOs.

ISE project in PROJECT folder, contain following files used to test IOs.

Io_test.bit	Bit file
Pin.ucf	UCF file
Io_test.vhd	VHDL file

PROCEDURE

- 1) Follow the procedure from step 18 by referring experiment no 1.
- 2) Select the given program **IO_TEST.bit** for testing Free IOs.
- 3) Download the program to target board
- 4) Check for DONE led which goes high indicating successful configuration.
- 5) Observe the output on CRO.
- 6) CRO should show a repetitive pattern as shown below on every pin from 1 to 24 of J6.



EXPERIMENT NO.3

EXPERIMENT NO.3

NAME

Study of counter design

OBJECTIVE

To write the program for 16 bit up counter and display on output LEDs.

EQUIPMENTS

VL-FPGA-B board and its power supply.

DESCRIPTION:

In this test/demo code we design a 16 bit up counter which is running at slower frequency and its output is mapped on the output LED's.

ISE project in 1_PROJECT folder, contain following files used to counter test.

counter.bit	Bit file
ucf_main.ucf	UCF file
counter.vhd	VHDL file

PROCEDURE

- 1) Follow the procedure from step 18 by referring experiment no 1.
- 2) Select the given program counter.bit for testing of designed counter with help of output LEDs.
- 3) Download the program to target board.
- 4) The LED's are glowing as the counter is running on them.

EXPERIMENT NO.4

EXPERIMENT NO.4

NAME

Study of interfacing of keyboard and display

OBJECTIVE

To write the program to interface 4 x 4 matrix keyboard and seven segment display. Also check the input switches and output LEDs.

EQUIPMENTS

- VL-FPGA-B board.
- Power Supply.
- J-Tag Cable..

DESCRIPTION:

VL-FPGA-B has a six multiplexed seven segment displays .Each individual character has a separate cathode control input.

To light an individual signal, drive the individual segment control signal High along with the associated cathode control signal for the individual character. The control signal is high, enabling the control inputs for the left-most character. The segment control inputs, A through G and DP, drive the individual segments that comprise the character. A High value lights the individual segment, a Low turns off the segment.

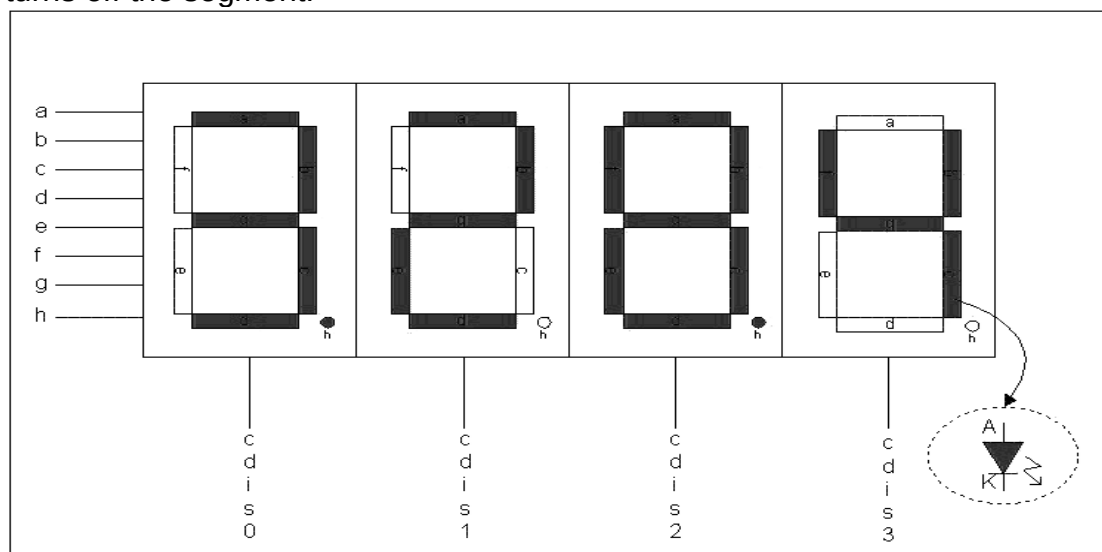
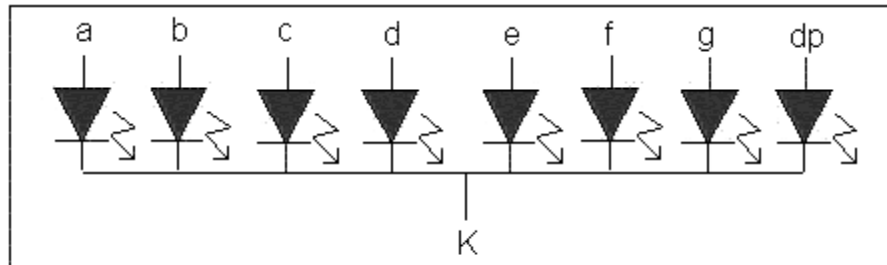


Figure 1: Seven Segment Display

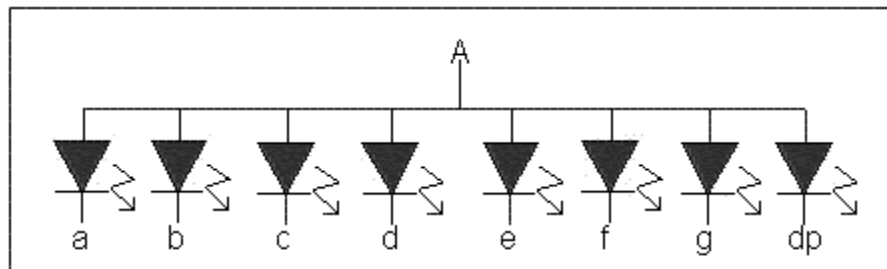
The two types of the seven segment displays are as shown below

- **Common Cathode Display:** In this type of display the cathode of all the LEDs are tied together and the anode terminals decides the status of the LED, either ON or OFF.
- To turn ON the LED i.e. segment value of driven segment should be 1 and 0 for turn OFF.



COMMON CATHODE DISPLAY

- **Common Anode Display:** In this type of display all the anode terminals of LEDs are tied together and the cathode terminals decide the status of the LED either ON or OFF.
- To turn ON the LED i.e. segment value of driven segment should be 0 and 1 for turn OFF.



COMMON ANODE DISPLAY

In this test/demo code we are checking the keyboard, LEDs, switches and seven segment display

ISE project in 1_PROJECT folder, contain following files used to test seven-segment display and keys.

keydis.bit	Bit file
pin.ucf	UCF file
key_display.vhd	VHDL file

PROCEDURE

- 1) Follow the procedure from step 18 by referring experiment no 1.
- 2) Select the given program keydis.bit for testing of designed counter with help of output LEDs.
- 3) Download the program to target board.
- 4) For DIP switch testing, as the switch (SW1 and SW2) is toggled from 0 to 1 the corresponding output LED (OL0 to OL15) glows.
- 5) For Key testing as the key is pressed its corresponding number will display on the seven segments. E.g. when key K0 is pressed, DIS 0 to DIS 5 displays 0. Similarly when key K15 is pressed, DIS 0 to DIS 5 displays F.

Table below shows key pressed and its response on seven segment display.

SR NO.	KEY PRESSED	DISPLAY
1	K0	0
2	K4	1
3	K8	2
4	K12	3
5	K1	4
6	K5	5
7	K9	6
8	K13	7
9	K2	8
10	K6	9
11	K10	A
12	K14	B
13	K3	C
14	K7	D
15	K11	E
16	K15	F

EXPERIMENT NO.5

EXPERIMENT NO.5

NAME

Study of LCD interfacing

OBJECTIVE

To write the program to interface LCD display to FPGA.

EQUIPMENTS

VL-FPGA-B board and its power supply.

DESCRIPTION:

VL-FPGA-B includes a LCD Module, which is a dot matrix liquid crystal display that displays alphanumeric, Kana (Japanese) characters and symbols. Built in controller provides connectivity between LCD and FPGA.

This LCD has a built in Dot Matrix controller, with font 5 X 7 or 5 X 10 dots, display data RAM for 80 characters (80 x 8 bit) and a character generator ROM which provides 160 characters with 5x7 font and 32 characters with font of 5x10. All the functions required for LCD are provided internally. Internal refresh is provided by the Controller.

The Interface details of the LCD display are as shown in figure 1.

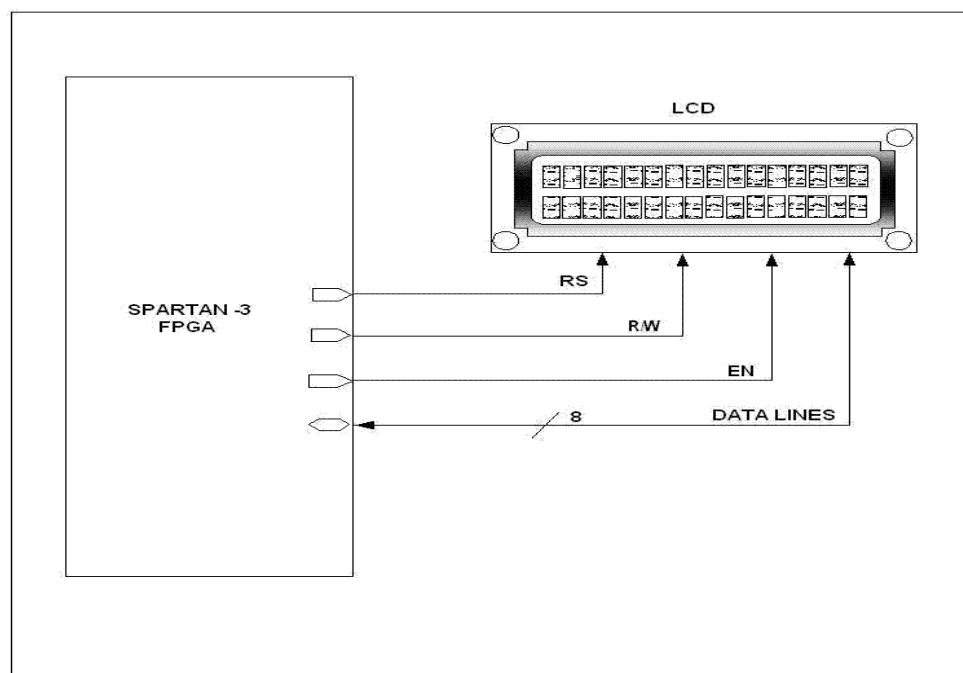


Figure 1: LCD Interface to SPARTAN-3 FPGA**DATA LINES CONNECTION**

LCD has 8 bit bidirectional data bus interface to FPGA. When Enable signal is at low level, this data bus remains in high impedance state. Interface details of the data lines with SPARTAN-3 FPGA are as in below Table

Data Line Interface to SPARTAN-3 FPGA

DATA BIT	FPGA Pin
"LCD_D<0>"	p167
"LCD_D<1>"	p166
"LCD_D<2>"	p165
"LCD_D<3>"	p162
"LCD_D<4>"	p161
"LCD_D<5>"	p156
"LCD_D<6>"	p155
"LCD_D<7>"	p154

CONTROL LINE INTERFACE:

The control lines of LCD comprises of RS, R/W# and E. The significance of the above mentioned control signals is as follows

- **RS:** Register select signal used to select Data register or a Command/Status register.
 - High on RS selects the data register.
 - Low on RS selects the Command/Status register.
- **R/W:** Read/Write select control line.
 - High on R/W # selects the read operation
 - Low on R/W # selects the write operation.
- **E:** Enable signal used to enable or disable the data bus.
 - Low on the enable signal puts the data bus into a high impedance state.
 - High on the enable signal selects the data bus

The control line interface of LCD with FPGA is as shown in below table

Control Line Interface to SPARTAN-3 FPGA

Control Bit	FPGA Pin
"LCD_E"	p168
"LCD_RS"	p171
"LCD_RW_BAR"	p169

Note: PR1 is used to adjust the contrast of LCD Display

ASCII CODE

The ASCII code for 5x 7 LCD Display is given in Figure 2

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
2	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL

Figure 2: ASCII Code for 5 x 7 LCD display

PROCEDURE

- 1) Follow the procedure from step 18 by referring experiment no 1.
- 2) Select the given program ***test_lcd_mod.bit***. for testing of designed counter with help of output LEDs.
- 3) Download the program to target board.
- 4) Check for DONE led which goes high indicating successful configuration
- 5) Press the RESET key.
- 6) LCD will display following pattern **"AKADEMIKA"**.

EXPERIMENT NO.6

APPENDIX A

Consolidated UCF for the Complete Board

CLOCK & RESET:

SR NO	SIGNAL NAME	PIN LOCATION
1	RESET	182
2	CLK_4M	181

4 X 4 MATRIX KEYBOARD:

SR NO	SIGNAL NAME	PIN LOCATION
1	RL0	95
2	RL1	94
3	RL2	93
4	RL3	81
5	SL0	85
6	SL1	86
7	SL2	87
8	SL3	88

LCD INTERFACE:

SR NO	SIGNAL NAME	PIN LOCATION
1	LCD_D0	167
2	LCD_D1	166
3	LCD_D2	165
4	LCD_D3	162
5	LCD_D4	161
6	LCD_D5	156
7	LCD_D6	155
8	LCD_D7	154
9	LCD_RS	171
10	LCD_R/W	169
11	LCD_E	168

SEVEN SEGMENT INTERFACE:

SR NO	SIGNAL NAME	PIN LOCATION
1	SEGA	144
2	SEGB	143
3	SEGC	141
4	SEGD	140

5	SEGE	139
6	SEGF	138
7	SEGG	137
8	SEGDP	135
9	DIS0	96
10	DIS1	97
11	DIS2	100
12	DIS3	101
13	DIS4	132
14	DIS5	133

INPUT SWITCHES:

SR NO	SIGNAL NAME	PIN LOCATION
1	SW1_8	57
2	SW1_7	52
3	SW1_6	51
4	SW1_5	50
5	SW1_4	48
6	SW1_3	46
7	SW1_2	45
8	SW1_1	44
9	SW2_8	43
10	SW2_7	42
11	SW2_6	40
12	SW2_5	39
13	SW2_4	37
14	SW2_3	36
15	SW2_2	35
16	SW2_1	34

OUTPUT LED's:

SR NO	SIGNAL NAME	PIN LOCATION
1	OL0	68
2	OL1	67
3	OL2	65
4	OL3	64
5	OL4	63

6	OL5	62
7	OL6	61
8	OL7	58
9	OL8	80
10	OL9	79
11	OL10	78
12	OL11	77
13	OL12	76
14	OL13	74
15	OL14	72
16	OL15	71

FREE IO's:

SR NO	SIGNAL NAME	PIN LOCATION
1	IO1	184
2	IO2	183
3	IO3	187
4	IO4	185
5	IO5	190
6	IO6	189
7	IO7	194
8	IO8	191
9	IO9	197
10	IO10	196
11	IO11	199
12	IO12	198
13	IO13	203
14	IO14	200
15	IO15	205
16	IO16	204
17	IO17	3
18	IO18	2
19	IO19	5
20	IO20	4
21	IO21	9
22	IO22	7
23	IO23	11
24	IO24	10
25	IO25	149
26	IO26	148
27	IO27	147
28	IO28	146

29	IO29	131
30	IO30	130
31	IO31	128
32	IO32	126
33	IO33	125
34	IO34	124
35	IO35	123
36	IO36	122
37	IO37	120
38	IO38	119
39	IO39	117
40	IO40	116
41	IO41	115
42	IO42	114
43	IO43	106
44	IO44	107
45	IO45	108
46	IO46	109
47	IO47	111
48	IO48	113
49	IO49	150
50	IO50	152

APPENDIX B

REFERENCES

Books on VHDL

- Fundamentals of Logic Design, by Charles Roth Jr
- Digital Systems Design using VHDL by Charles Roth Jr,
- VHDL by Douglas Perry,
- The Designers Guide to VHDL by Peter Ashenden.
- VHDL Primer by Jayram Bhaskar.
- XILINX data book, 2003

Books on Digital Design

- Digital Design Principles and Practices by John F. Wakerly
- Digital design by Morris Mano.
- An Engineering Approach to Digital Designs by William Fletcher.
- Digital Logic and computer Design by Morris Mano.
- Computer Organization by Hamacher, Vranesic and Zaky.
- Digital System Design and Microprocessors by John P. Hayes.
- Computer Architecture and Organization by John P. Hayes.

Books on Verilog

- Verilog HDL- A Guide To Digital Design & Synthesis By Sameer Palnitkar
- A Verilog HDL Primer By J. Bhaskar
- A Verilog HDL Synthesis-A Practical Primer By J. Bhaskar

Books on CMOS VLSI

- Principles Of CMOS VLSI Design 2e By Neil H. E. Weste, Kamran Eshraghian
- Digital Integrated Circuits: A Design Perspective by J. Rabaey.
- CMOS Digital Integrated Circuits by Pucknell and Eshraghian.
- Principles of CMOS VLSI Design by Pucknell and Eshraghian

APPENDIX C:

GLOSSARY

ASIC - APPLICATION SPECIFIC INTEGRATED CIRCUIT

The common name for semi-custom integrated circuits. These can include digital, linear and mixed-level circuits.

BASIC CELL

The smallest physical area on an ASIC array repeated in both the X and Y direction to populate the core matrix. It contains unconnected components that can be programmed individually or in-groups to form macros of varying complexity.

BUILT IN SELF TEST (BIST)

The capability of a product to carry out a functional test of itself. Some support from external equipment may be required. BIST usually involves special hardware in the product to generate input stimuli and to analyze test responses.

CPLD - COMPLEX PROGRAMMABLE LOGIC DEVICES

CPLD's are made up of smaller common macro cells, which are programmable. These are EPROM based devices i.e. their configuration.

CONFIGURATION

Configuration is the process of loading design specific programming data into one or more FPGA's to define the functional operation of the internal blocks and their interconnections.

DESIGN FOR TESTABILITY (DFT)

A strategy to test your ASIC defined at the start of the design process rather than at the end.

EDIF - ELECTRONIC DESIGN INTERCHANGE FORMAT.

A textual language designed to allow all forms of electronic design information transfer between different CAD systems. Currently implemented for netlist and schematic circuit descriptions, although there are significant differences between different tool vendor implementations (which may impair portability).

FPGA - FIELD PROGRAMMABLE GATE ARRAY

They are arrays of logic blocks, which can be interconnected to form complex logic implementation.

They are RAM based devices.

GATE COUNT

A measure of the size of a design, usually expressed in terms of the equivalent number of basic 2 input nand gates required.

JTAG - Joint Test Action Group

This group created the foundation for the IEEE work, to set specifications, that enables board and chip level functional verification of a board during production.

It was the Committee that established the Test Access Port (TAP) and boundary-scan architecture defined in IEEE Standard 1149.1-1990.

LIBRARY

(ASIC library) Set of macros offered by ASIC vendor. Contains hard macro gates (for example, 2-input NAND gate); can also contain soft macro functions (for example, 74161 counter) for improved design productivity.

NETLIST

(ASIC usage) A representation of an ASIC design as a set of library cells and their interconnections. Frequently, but not necessarily, a text file or set of text files.

RTL -REGISTER TRANSFER LEVEL.

A subset of behavioural modelling constructs, which can be used to model a circuit at the level of data flowing between a set of registers. This level of abstraction typically contains little timing information, except references to a set of clock edges and features.

SPROM

Serial Programmable Read Only Memory

UCF - USER CONSTRAINT FILE

This is a file where the user has to give the constraints of pin locking according to the Trainer hardware.

Pin locking is - assigning the signals of the code to specific pins of Target Device.

VHDL - VERY HIGH SPEED INTEGRATED CIRCUIT HARDWARE DESCRIPTION LANGUAGE

Developed under guidelines from the U.S. Department of Defence and adapted as IEEE Standard 1076.

XNF - Xilinx Netlist Format.